



**Scientific / Metrology Instruments**  
**Electron Microscope**

Solutions for Innovation

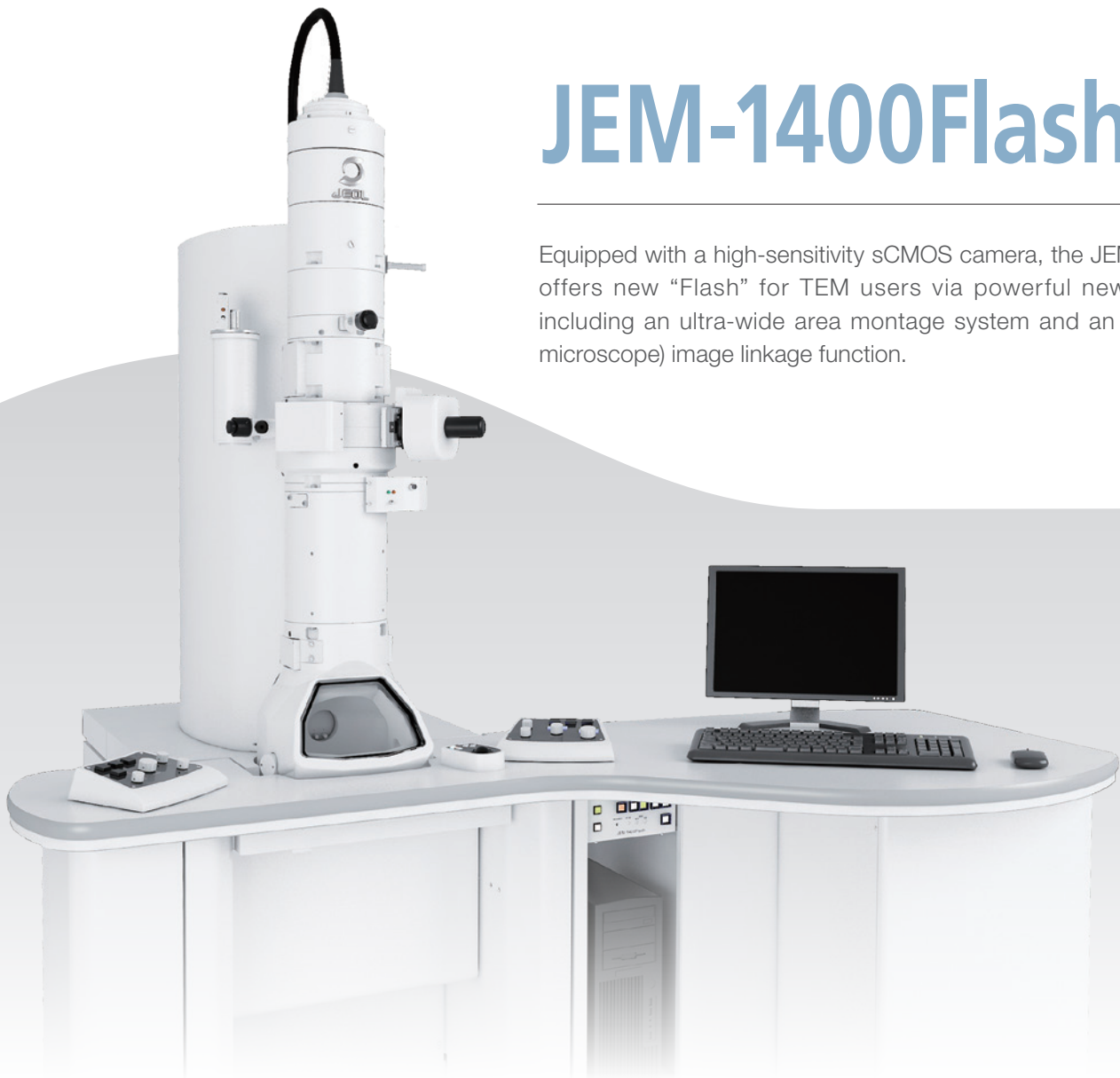
# JEM-1400Flash



JEOL Ltd.

# JEM-1400Flash

Equipped with a high-sensitivity sCMOS camera, the JEM-1400Flash offers new “Flash” for TEM users via powerful new functions, including an ultra-wide area montage system and an OM (optical microscope) image linkage function.

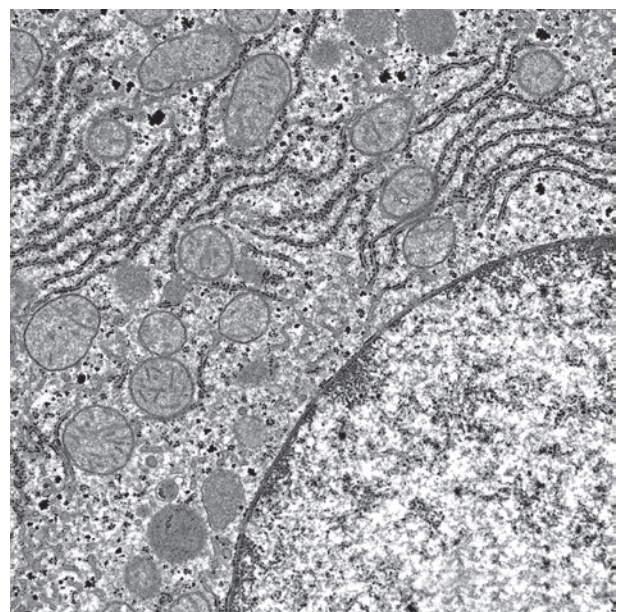


## JEOL's high-sensitivity sCMOS camera “Matataki Flash” camera

>>>The JEM-1400Flash comes with a high-sensitivity sCMOS camera. “Matataki Flash”, JEOL's innovative sCMOS camera, dramatically reduces the readout noise while possessing high frame rate comparable to that of TV. This powerful feature enables the acquisition of sharp TEM images with extremely low-noise.

### Main specifications of sCMOS camera for JEM-1400Flash

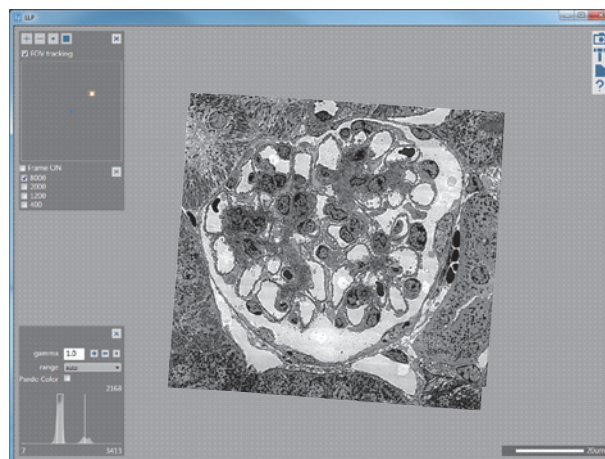
|                      |                               |
|----------------------|-------------------------------|
| Readout noise        | 0.8 electrons                 |
| Frame rate           | 30 fps (unbinned)             |
| Pixels               | 2048 x 2048 pixels            |
| Saved image format   | 16 bit TIFF, BMP, JPEG        |
| Effective pixel size | 21.7 x 21.7 (μm) <sup>2</sup> |



Specimen: Rat liver    Accelerating voltage: 120 kV    1.0μm

## Ultra-wide area montage system Limitless Panorama (LLP)

>>In addition to the conventional electromagnetic image shift, the JEM-1400Flash comes with a montage system capable of utilizing stage drive for the field shift. This new system allows for simple capture of a montage panorama image over a limitless wide area. Thus, an ultra-wide area, high pixel-resolution image is obtainable, which is comparable to the image taken by conventional photo film. The captured image can be saved by cutting from any area and in any direction. The saved image is subjected to size measurement by displaying it on the TEM center software or the offline Viewer software "Sight Viewer".



Operation screen of LLP

20µm

LLP basic  
operation →



LLP application  
data →



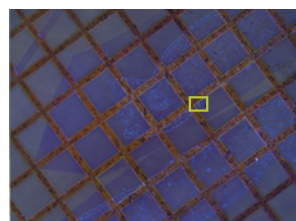
## OM image linkage function Picture Overlay

>>A digital image acquired with an OM can be overlaid on a TEM image. Since an observation area can be searched on the overlaid image, high resolution observation of a fluorescence site is easily made on the TEM image. Combined use of the TEM control software (TEM Center) and the Stage Navigation system enables fast specimen movement to the target area.

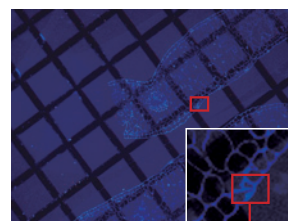
Picture Overlay basic operation →



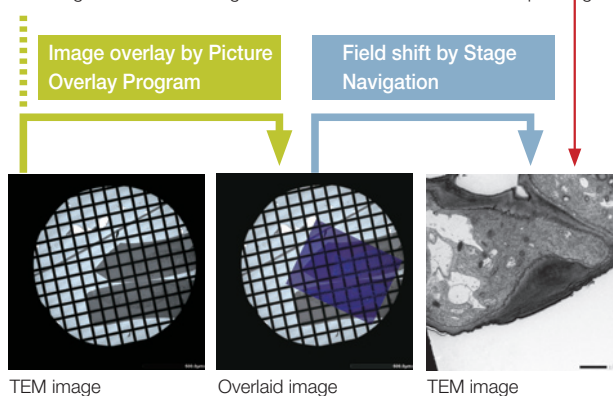
### Basic operation flow of Picture Overlay



OM image and overlaid image



Fluorescence microscope image



TEM image  
(Ultra lowmag)

Overlaid image

TEM image

Specimen: Leaf of spinach

## Main Performance and Basic Configuration of JEM-1400Flash

|                                                      | JEM-1400Flash HC  | JEM-1400Flash HR  |
|------------------------------------------------------|-------------------|-------------------|
| TEM resolution                                       | 0.2 nm            | 0.14 nm           |
| Accelerating voltage                                 | 10 to 120 kV      |                   |
| Magnification                                        | ×10 to ×1,200,000 | ×10 to ×1,500,000 |
| Maximum tilt angle(Tilt-X) *1                        | ± 70°             | ± 70°             |
| Maximum number of samples to load*2                  | 4                 |                   |
| Vacuum system                                        | TMP               |                   |
| JEOL's sCMOS camera                                  | ○                 |                   |
| Ultra-wide area montage system<br>LimitLess Panorama | ○                 |                   |
| OM image linkage<br>Picture Overlay                  | ○                 |                   |
| Use of special holders                               | ○                 |                   |

\*1 With optional high tilt specimen holder, EM-11230HTH

\*2 With optional specimen quartet holder, EM-01070SQH

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\* Improvements or changes to device appearance or specifications may be made without notice.



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